# Data Cleaning Methodology

The cleaning process was designed using a rule-based remediation framework derived from the data profiling findings and Data Quality Rulebook. Each transformation was implemented to address specific data quality issues identified during profiling.

## Cleaning Steps Performed

**1. Duplicate Record Removal**

Removed duplicate records based on business attributes while excluding the unique_key field. Where duplicates existed, the most recent record was retained.

**2. Invalid Date Sequence Removal**

Removed records where:

- created_date > closed_date
- created_date > due_date

These records violated temporal integrity rules defined in the Data Quality Rulebook.

**3. Future Date Validation**

Removed records where:

- closed_date > '2026-06-03'

These records represented future dated service requests and violated timeliness requirements.

**4. Status Reconciliation**

Updated status values to 'Closed' where:

- closed_date IS NOT NULL
- status <> 'Closed'

This ensured consistency between workflow status and closure information.

**5. Closed Status Validation**

Removed records where:

- status = 'Closed'
- closed_date IS NULL

These records violated the closed request lifecycle rules.

**6. City Standardization**

Standardized city values using a predefined mapping table to consolidate naming variations into approved formats.

Examples:

- NYC → NEW YORK

- Ny → NEW YORK

**7. Borough Standardization**

Standardized borough values using a predefined mapping table.

Examples:

- Brooklyn → BROOKLYN
- Brooklyn NY → BROOKLYN

**8. Complaint Type Standardization**

Standardized complaint categories using a mapping table to eliminate naming inconsistencies while preserving business meaning.

**9. ZIP Code Validation**

Removed records containing invalid ZIP codes while preserving records with NULL ZIP values.

**10. Data Type Conversion**

Converted ingestion layer VARCHAR columns to appropriate production datatypes based on the data dictionary.

Examples:

- DATETIME
- INT
- DECIMAL
- VARCHAR

**11. Duplicate Verification**

Performed duplicate validation checks following all cleaning activities to ensure duplicate records had not been introduced during transformation.

**12. Due Date Column Assessment**

Removed the due_date column from the analytical layer due to insufficient population and limited analytical value.

**13. Post Transformation Duplicate Verification**

Performed a second duplicate validation following schema modifications.

**14. ZIP Code Revalidation**

Performed final ZIP code validation checks to ensure compliance with data quality standards after all transformations were completed.

**Exception Management**

All removed records were logged to an exception table containing:

- unique_key

- exception reason

This preserved lineage and enabled traceability back to the Bronze layer, ensuring that no records were permanently lost without an audit trail.

**Outcome**

The cleaning process successfully addressed issues related to duplication, temporal validity, status consistency, categorical standardization, ZIP code quality, and datatype enforcement while maintaining transparency through exception management and validation reporting.